

Ho Fung Tsoi

Postdoctoral Researcher

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Last updated: Sep 2026

Research Interests

Building machine learning methods for accelerating scientific discoveries at particle physics experiments.

Experimental particle physics: new physics searches (SUSY, exotic Higgs) and trigger system at the CERN LHC

Machine learning: self-supervised pre-training methods, anomaly detection, symbolic regression

Fast inference on FPGAs: low-latency (sub-microsecond) machine learning algorithms on FPGAs

Education

University of Wisconsin–Madison, Ph.D. Physics 2024

Thesis: [Search for exotic Higgs boson decays with CMS and fast machine learning solutions for the LHC](#)

Advisor: Sridhara Dasu

The Chinese University of Hong Kong, B.Sc. Physics 2018

Thesis: First passage time problem of the time-dependent Ornstein-Uhlenbeck process

Advisor: Chi-Fai Lo

Visiting student at University of California Berkeley & Berkeley Lab (Jan – Aug 2017)

Professional Experience

Postdoctoral Researcher, University of Pennsylvania (Supervisor: Dylan Rankin) 2024 – Present

Awards and Honors

CERN, [CMS PhD Thesis Award](#) 2024

UWisconsin, Cornelius P. and Cynthia C. Browne Fellowship 2024

CUHK, CN Yang Scholarship 2017

CUHK, Wei Lun Foundation Exchange Scholarship 2017

CUHK, Professor Dennis Yam Kuen Lo Physics Award 2016

Selected Research Experience

Search for new physics at the LHC

- **SUSY.** ATLAS Run 3 search for electroweak production of supersymmetric particles in compressed mass spectra with low-momentum tracks (2026–Present).
- **Exotic Higgs.** CMS Run 2 search for exotic Higgs boson decays into pseudoscalars in the final state of two b quarks and two τ leptons, and the combination with the $\mu\mu b\bar{b}$ final state [4] (2020–2024).

Novel ML Methods for Particle Physics Analyses

- **Self-supervised learning.** Self-distillation pre-training method **jBOT** for jets [14]; VICReg pre-training to improve resonance mass width reconstruction; data-driven formation of invariant training pairs using energy mover's distance (2025–Present).
- **SymbolFit.** Automatic discovery of parametric functions to model HEP data [6,13] (2024–2026).

Low-Latency (Sub-Microsecond) ML Algorithms on FPGAs

- **hls4ml.** Co-developer; an open-source tool that converts and optimizes trained neural networks into FPGA firmware [7] (2023–Present).
- **SpikePID.** Neuromorphic spiking neural networks (SNNs) for on-detector dN/dx clustering counting in future drift chambers [11] (2025–Present).
- **SparsePixels.** CNNs with sparse convolution for spatially sparse image data on FPGAs [8] (2024–Present).
- **SymbolNet.** Compression of neural networks using symbolic regression for fast inference [3,5] (2022–2024).

Trigger System at the LHC

- **GNTau.** Commissioning the GNN-based tau trigger at the ATLAS High-Level Trigger for Run 3 [9] (2024–2025).

- **CICADA**. Development of an anomaly detection trigger algorithm at the CMS Level-1 Trigger [10] (2022–2024).
- Development of the Data Quality Monitoring (DQM) software for the Calorimeter Layer-1 subsystem at the CMS Level-1 Trigger (2020–2024).
- HL-LHC trigger sensitivity projection for the CMS searches for exotic Higgs boson decays [1] (2020–2021).

Selected Publications

Listed below are selected publications, conference proceedings, and technical reports to which I was a primary author (*: refereed). I also co-authored papers by [CMS](#) and [ATLAS](#) as a collaboration member. For the complete list see [INSPIRE](#). Preprints at [arXiv](#).

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|-------|---|------|
| [14*] | “jBOT: Semantic Jet Representation Clustering Emerges from Self-Distillation”
Ho Fung Tsoi , Dylan Rankin
SciPost Phys. 21 , 053 (2026) [arXiv:2601.11719] | 2026 |
| [13] | “Machine Can Automatically Discover Parametric Functions to Model HEP Data”
Ho Fung Tsoi , Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles
Proceedings of ICHEP 2026 [arXiv:2607.19750] | 2026 |
| [12] | “SparsePixels++: Scalable Sparse Convolution on FPGAs”
Ho Fung Tsoi , Dylan Rankin, Philip Harris
<i>manuscript in preparation for submission</i> | 2026 |
| [11] | “SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers”
Ho Fung Tsoi , Kam Wai Lai, Dylan Rankin
<i>manuscript in preparation for submission</i> | 2026 |
| [10] | “Unsupervised Anomaly Detection for Real-Time Data Acquisition in the CMS Experiment at the LHC”
CMS Collaboration
<i>manuscript in preparation for submission</i> [CMS-PAS-MLG-25-001] | 2026 |
| [9] | “Performance of the ATLAS Tau Trigger in Run 3”
ATLAS Collaboration
<i>manuscript in preparation for submission</i> | 2026 |
| [8*] | “SparsePixels: Efficient Convolution for Sparse Data on FPGAs”
Ho Fung Tsoi , Dylan Rankin, Vladimir Loncar, Philip Harris
accepted to Mach. Learn.: Sci. Technol. [arXiv:2512.06208] | 2025 |
| [7*] | “hls4ml: A Flexible, Open-Source Platform for Deep Learning Acceleration on Reconfigurable Hardware”
Jan-Frederik Schulte et al.
ACM Trans. Reconf. Tech. Syst. 19 , 2 (2026) [arXiv:2512.01463] | 2025 |
| [6*] | “SymbolFit: Automatic Parametric Modeling with Symbolic Regression”
Ho Fung Tsoi , Dylan Rankin, Cecile Caillol, Miles Cranmer, Sridhara Dasu, Javier Duarte, Philip Harris, Elliot Lipeles, Vladimir Loncar
Comput Softw Big Sci 9 , 12 (2025) [arXiv:2411.09851] | 2024 |
| [5*] | “SymbolNet: Neural Symbolic Regression with Adaptive Dynamic Pruning for Compression”
Ho Fung Tsoi , Vladimir Loncar, Sridhara Dasu, Philip Harris
Mach. Learn.: Sci. Technol. 6 , 015021 (2025) [arXiv:2401.09949] | 2024 |
| [4*] | “Search for Exotic Decays of the Higgs Boson to a Pair of Pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ Final States”
CMS Collaboration
Eur. Phys. J. C 84 , 493 (2024) [arXiv:2402.13358] | 2024 |
| [3*] | “Symbolic Regression on FPGAs for Fast Machine Learning Inference”
Ho Fung Tsoi , Adrian Alan Pol, Vladimir Loncar, Ekaterina Govorkova, Miles Cranmer, Sridhara Dasu, Peter Elmer, Philip Harris, Isobel Ojalvo, Maurizio Pierini
EPJ Web of Conferences 295 , 09036 (2024) [arXiv:2305.04099] | 2023 |

- [2] “Searches for exotic Higgs boson decays with the CMS experiment” 2023
[Ho Fung Tsoi](#)
[Proceedings of the European Physical Society Conference on High Energy Physics \(PoS EPS-HEP2023 402\)](#)
- [1] “The Phase-2 Upgrade of the CMS Data Acquisition and High Level Trigger” 2021
 CMS Collaboration
[CMS Technical Design Report \(2021\)](#), [CERN-LHCC-2021-007](#), [CMS-TDR-022](#)

Conference, Workshop, Seminar Presentations

- [13] Fast Machine Learning for Science Conference (FastML) — UCSD, USA Sep 2026
[SpikePID: Neuromorphic Cluster Counting in Future Drift Chambers](#)
[SparsePixels++: Scalable Sparse Convolution on FPGAs](#)
- [12] International Conference on High Energy Physics (ICHEP) — Natal, Brazil Jul 2026
[Machine Can Automatically Discover Parametric Functions to Model HEP Data](#)
- [11] TREASURE: Tokenizing HEP Collider Data for AI Discovery — BNL, USA Apr 2026
[Self-Supervised Learning: Machine Can Learn a Lot by Just Observing](#)
- [10] Coordinating Panel on Advanced Detectors (CPAD) — UPenn, USA Oct 2025
[SparsePixels: Efficient Convolution for Sparse Data on FPGAs](#)
- [9] 2024 CMS Thesis Award Ceremony Talk (CMS Week) — CERN, Switzerland Oct 2025
[Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC](#)
- [8] Fast Machine Learning for Science Conference (FastML) — ETH Zurich, Switzerland Sep 2025
[SparsePixels: Efficient Convolution for Sparse Data on FPGAs](#)
- [7] High-energy physics seminar — UPenn; U.Washington; Fermilab, USA 2023, 2024
[Search for Exotic Higgs Boson Decays with CMS and Fast Machine Learning Solutions for the LHC](#)
- [6] US LHC Users Association Meeting (US LUA) — Fermilab, USA Dec 2023
[CICADA: Anomaly Detection for New Physics Searches at the CMS Level-1 Trigger](#)
- [5] Machine Learning at Level-1 Trigger Workshop (ML@L1) — CERN, Switzerland Dec 2023
 Anomaly Detection – CICADA: Status, Plans, and Prospects for Phase-2
- [4] CMS Machine Learning Town Hall — CERN, Switzerland Sep 2023
 L1 Anomaly Detection with Calorimeter Inputs: Status and Opportunities
- [3] European Physical Society Conference on High Energy Physics (EPS-HEP) — Hamburg, Germany Aug 2023
[Searches for exotic Higgs boson decays with the CMS experiment](#)
- [2] International Conference on Computing in High Energy & Nuclear Physics (CHEP) — Norfolk VA, USA May 2023
[Symbolic Regression on FPGAs for Fast Machine Learning Inference](#)
- [1] International Conference on Applied Mathematics — CityU, Hong Kong May 2016
 First Passage Time Problem of the Time-Dependent Ornstein-Uhlenbeck Process: a Model for Stochastic Decision-Making Process

Professional Services

Membership

- Member, [ATLAS Collaboration](#), CERN 2024 – Present
- Affiliated member, [Accelerated AI Algorithms for Data-Driven Discovery \(A3D3\) Institute](#) 2024 – Present
- Member, [CMS Collaboration](#), CERN 2020 – 2024

Collaboration Role

- **Co-convener**, SUSY Monte Carlo & Interpretation subgroup [CMS] 2023 – 2024
- **Contact person**, Higgs boson Monte Carlo group [CMS] 2021 – 2023
- **Software developer**, Data Quality Monitoring system for CaloLayer-1 trigger [CMS] 2020 – 2024

Journal Referee

Transactions on Machine Learning Research (2026); *IEEE Transactions on Evolutionary Computation* (2026); *Wiley Advanced Theory and Simulations* (2025).

Teaching and Mentoring

Teaching

- Guest lecturer [UPenn], Data Analysis for the Natural Sciences II: Machine Learning (PHYS 3359, Spring 2025)
- Teaching assistant [U.Wisconsin], undergraduate physics courses (Fall 2018 – Spring 2020)

Master students

- Kam Wai Lai [UPenn]: Spiking neural network on FPGAs for cluster counting in drift chambers (2025–2026)
- Yuyan Wang [UPenn]: Joint-Embedding Predictive Architecture (JEPAs) for jet anomaly detection (2025–2026)

Undergraduate students

- Chenjia Ni [UPenn]: Lorentz-equivariant architecture on FPGAs (2026–Present)
- Abhay Agarwal [UPenn]: Scalable sparse convolution on FPGAs (2026–Present)
- Alex Yang [UPenn]: Mass regression of supersymmetric particles with contrastive learning (2025)
- Ashni Kumar [Drexel]: Acceleration of reinforcement learning agent inference on FPGAs for astronomical observations (2025)